

Heaven's light is our guide



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Project Report Book

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Course Title: Power Electronics and Drives Sessional

Project Title: Design of Boost Converter from 9V to 36V.

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Chapter 1

Applying Power Electronics Knowledge

1.1 Introduction

A boost converter is a type of DC-to-DC converter that increases a low input voltage to a higher output voltage. Its operation is based on the energy-storing property of an inductor. The circuit consists of a single inductor, a controlled switch (MOSFET), a rectifier (diode), and a capacitor. The conversion process is divided into two main stages, determined by the state of the switch.

1. **Switch ON** : When the switch is closed, the input voltage is applied across the inductor. Current flows through the inductor, causing it to store energy in its magnetic field. The diode is reverse-biased, isolating the output from the input, and the output capacitor supplies power to the load.
2. **Switch OFF** : When the switch is opened, the inductor's magnetic field collapses, and it releases the stored energy. To maintain current flow, the inductor's voltage reverses polarity. This released energy, along with the input voltage, forward-biases the diode, sending a combined, higher voltage to the output capacitor and the load.

In continuous-conduction mode, the ideal static conversion ratio is

$$\frac{V_{out}}{V_{in}} = \frac{1}{(1 - D)}$$

where D represents the duty cycle. Control, component ratings, and layout are crucial for efficient and stable operation [1][3].

Step-up ratio calculation for the 9 V to 36 V target

- Target step-up ratio:

$$\frac{V_{out}}{V_{in}} = \frac{36}{9} = 4$$

a 4:1 step-up.

- Ideal duty cycle at 9 V input:

$$D = 1 - \frac{V_{out}}{V_{in}} = 1 - \frac{9}{36} = 1 - 0.25 = 0.75$$

Current-mode control with slope compensation is used to maintain stable operation at duty cycles above 50%. Device ratings, including the switch, diode/synchronous FET, inductor, and capacitors, along with thermal margins, are selected to handle the higher voltages and currents involved in this conversion [2]. Core sizing equations for the inductor, input/output capacitors, sense and feedback resistors, as well as ripple and ripple-factor calculations, help set initial values and define performance targets before making adjustments [3].

Beyond the power stage and control loop, the design focuses on electromagnetic compatibility and practical manufacturability. High di/dt loop areas are minimized through careful PCB layout. Snubbers and clamps help reduce ringing. Conducted and radiated emissions are managed with filtering and measurement best practices.

1.2 Principle of Operation

The operation of the boost converter is governed by the fundamental energy storage and release properties of an inductor. The circuit typically consists of four essential components:

1. An inductor (L) for energy storage.
2. A MOSFET (switch) for controlled energy transfer.
3. A diode (D) to enforce unidirectional current flow.
4. A capacitor (C) for voltage smoothing and load support.

The working cycle can be explained in two modes:

- **Switch ON phase:** When the MOSFET is turned ON, the input voltage is directly applied across the inductor. The inductor resists the sudden change in current, resulting in a gradual buildup of current and magnetic energy. During this period, the diode is reverse biased, isolating the load from the input. The load is powered solely by the capacitor.
- **Switch OFF phase:** When the MOSFET is turned OFF, the inductor's magnetic field collapses, releasing its stored energy. The polarity of the induced voltage reverses, forward biasing the diode. The inductor current, combined with the source voltage, charges the capacitor and supplies the load. The result is an output voltage greater than the input voltage.

1.3 Mathematical Relation

In continuous conduction mode (CCM), where the inductor current never falls to zero, the steady-state voltage conversion ratio is given as:

$$\frac{V_{out}}{V_{in}} = \frac{1}{1 - D}$$

where:

- V_{in} = input voltage
- V_{out} = output voltage
- D = duty cycle (fraction of time switch is ON)

This equation shows that as the duty cycle increases, the output voltage rises. For instance, at a duty cycle of 0.75 (75%), the output theoretically reaches four times the input. However, in practical systems, factors such as diode forward voltage drop, MOSFET on-resistance, and inductor resistance reduce the achievable output.

At higher duty cycles (>50%), stability becomes critical, requiring current-mode control with slope compensation to prevent subharmonic oscillations.

Application of Knowledge in Design

The successful design of this project required the integration of multiple aspects of power electronics theory into practical engineering decisions:

- **Inductor and Capacitor Dynamics:** Knowledge of inductor current ripple and capacitor voltage ripple directly guided component sizing. By carefully selecting the inductance and capacitance, we ensured continuous conduction and limited output voltage ripple to within 1% of the rated voltage.
- **Semiconductor Device Selection:** Understanding the role of switching devices was essential in choosing an appropriate MOSFET with low $R_{ds(on)}$ to minimize conduction losses, and a Schottky diode with low forward voltage drop to enhance efficiency.
- **Thermal and EMI Considerations:** Familiarity with electromagnetic interference (EMI) sources and thermal effects allowed us to select switching frequencies and component ratings that balance efficiency with safe operation. Proper PCB layout practices, such as minimizing loop areas and providing adequate heat dissipation, translated theoretical knowledge into a robust prototype.

1.4 Broader Implications

This project highlights how the application of core power electronics knowledge—from circuit theory to practical component behavior—enables the realization of efficient, low-cost, and scalable energy conversion solutions. It illustrates the link between theoretical analysis and practical implementation, ensuring that abstract design equations become a working hardware system capable of real-world performance.

Thus, the development of the 9 V to 36 V boost converter demonstrates not only the mastery of fundamental concepts but also the capacity to apply them meaningfully in engineering practice.

Chapter 2

Designing Engineering Solutions

Engineering design is the process of translating theoretical principles into practical, reliable, and optimized solutions. In this project, the goal was to design a boost converter capable of stepping up an input voltage of 9 V to a regulated 36 V output, operating at a switching frequency of 32 kHz and supplying a 0.5 A load current.

The design process involved a structured approach: defining specifications, performing analytical calculations, selecting appropriate components, developing a control algorithm, and validating the design through hardware implementation.

2.1 Boost Converter Design Calculations ($f = 32 \text{ kHz}$)

1) System Specifications

Input Voltage, V_{in}	9 V
Output Voltage, V_{out}	36 V
Switching Frequency, f	32 kHz
Assumed Load Current, I_{out}	0.5 A
MOSFET	IRFZ44N (N-channel)
PWM / Feedback	Arduino NANO (Timer2, Pin 11) with voltage divider feedback

Component shortlist used in the prototype: Arduino UNO $\times 1$, IRFZ44N $\times 1$, Schottky diode (see note), Inductor $\approx 300\text{--}350 \mu\text{H}$, Output Capacitor 50 V 470 μF $\times 1$, Resistors 10 k Ω & 90 k Ω , Pins/Headers, Screw Terminal, Wires, PCB.

2) Key Design Targets

- Continuous Conduction Mode (CCM) operation at the nominal load.
- Output voltage ripple $\leq 1\%$ ($\approx 0.36 \text{ V}$) as a working target.
- Input ripple modest ($\leq 0.3 \text{ V}$) to reduce source stress.
- Practical component ratings with $\geq 20\text{--}50\%$ margin where possible.

3) Duty Cycle (D)

Ideal boost relation:

$$V_{out} = \frac{V_{in}}{(1 - D)}$$

$$\Rightarrow D_{ideal} = 1 - \frac{V_{in}}{V_{out}}$$

Using $V_{in} = 9 \text{ V}$ and $V_{out} = 36 \text{ V}$: $D_{ideal} = 1 - 9/36 = 0.75$ (75%).

Including approximate drops (Schottky $\sim 0.4 \text{ V}$, switch $\sim 0.2 \text{ V}$):

$$D \approx 1 - \frac{V_{in} - V_{switch}}{V_{out} + V_{diode}} = 1 - \frac{9 - 0.2}{36 + 0.4} = 0.758 = 75.8\%$$

For $V_{out} = 30\text{V}$: $D = 1 - 9/30 = 0.7$ (70%)

For $V_{out} = 40V$: $D = 1 - 9/40 = 0.775$ (77.5%) This corresponds to PWM values of approximately 179 to 198 (since analogWrite uses 0-255).

4) Inductor Design

The inductance ripple current (ΔI_L) is calculated as :

$$\Delta I_L = \frac{V_{in} \times D}{f \times L}$$

Average inductor current in CCM equals input current: $I_{L(avg)} = I_{in} = \frac{I_{out}}{1-D}$

For $V_{out} = 40V$ ($D = 0.775$): $\Delta I_{L(avg)} = \frac{1}{0.225} \approx 4.44A$

For $V_{out} = 30V$ ($D = 0.7$): $\Delta I_{L(avg)} = \frac{1}{0.3} \approx 3.33A$

A typical design sets ΔI_L to 20-40% of $I_{L(avg)}$.

Using 30% at $V_{out} = 40V$: $\Delta I_L = 0.3 \times 4.44 \approx 1.33A$

For simplicity, we target $\Delta I_L = 1A$. Rearranging for L:

$$L = \frac{V_{in} \times D}{f \times \Delta I_L}$$

$V_{in} = 9V$, $D = 0.775$, $f = 62500$ Hz, $\Delta I_L = 1A$:

$$L = \frac{9 \times 0.775}{62500 \times 1} = \frac{6.975}{62500} \approx 0.1116mH = 111.6\mu H$$

Choosing a standard value, $L = 100\mu H$ is appropriate. Verify the ripple:

$$\Delta I_L = \frac{9 \times 0.775}{62500 \times 100 \times 10^{-6}}$$

This is $\approx 25\%$ of I_{Lavg} , which is acceptable. The peak inductor current is:

$$I_{L(peak)} = I_{Lavg} + 2\Delta I_L = 4.44 + 2 \times 1.12 \approx 5.0A$$

Ensure the inductor is rated for at least 5A

5) Capacitance Calculation

The output voltage ripple (ΔV_{out}) is approximated as :

$$\Delta V_{out} = \frac{I_{out} \times D}{f \times C}$$

Target $\Delta V_{out} = 0.4V$ (1% of 40V) :

$$C = \frac{I_{out} \times D}{f \times \Delta V_{out}}$$

$I_{out} = 1A$, $D=0.775$, $f=62500Hz$, $\Delta V_{out} = 0.4V$:

$$C = \frac{1 \times 0.775}{62500 \times 0.4} = \frac{0.775}{25000} \approx 31\mu F$$

A larger capacitor improves stability, using $C=100\mu F$:

$$\Delta V_{out} = \frac{1 \times 0.775}{62500 \times 100 \times 10^{-6}} \approx 0.124V$$

This is $\sim 0.31\%$ of $40V$, well below 1% . The capacitor should be rated for at least $50V$.

6) Voltage Divider Design

The feedback voltage (V_{fb}) must be scaled to $0-5V$ for the Arduino's ADC:

- $V_{out} = 30V \rightarrow V_{fb} \approx 3V$ ($analogRead \approx 614$)
- $V_{out} = 40V \rightarrow V_{fb} \approx 4V$ ($analogRead \approx 818$) using:

$$V_{fb} = V_{out} \times \frac{R_2}{R_1 + R_2}$$

Set $V_{fb} = 4V$ at $V_{out} = 40V$:

$$\frac{R_2}{R_1 + R_2} = \frac{4}{40} = 0.1$$

$$\rightarrow R_1 = 9R_2$$

Choose $R_2 = 10k\Omega$, $R_1 = 90k\Omega$. Verify:

- $V_{out} = 40V$: $V_{fb} = 40 \times \frac{10}{100} = 4V$
- $V_{out} = 30V$: $V_{fb} = 30 \times \frac{10}{100} = 3V$

2.2 Algorithm

The control loop operated continuously. At each step, the feedback voltage was measured and compared with the desired reference. If the actual feedback voltage was lower than expected, the PWM duty cycle was incrementally increased, constrained within safe limits. Conversely, if the actual feedback was higher than expected, the duty cycle was decreased accordingly. When the actual and desired voltages were equal, the duty cycle remained unchanged. The updated PWM value was then applied to the switching MOSFET. This closed-loop adjustment ensured stable regulation of the output voltage.

The algorithm for the control of the pulse width modulation of the boost converter can be described as:

1. **Initialization:**
 - 1.1 Set feedback as INPUT.
 - 1.2 Set PWM as OUTPUT.
 - 1.3 Configure PWM frequency.
 - 1.4 Initialize pwm with starting duty cycle.

2. **Control Loop (repeat indefinitely):**
 - 2.1 Read $\text{actualFeedback} \leftarrow \text{analogRead}(\text{feedback})$.
 - 2.2 If $\text{desiredFeedback} > \text{actualFeedback}$:
 - a) Increase $\text{pwm} \leftarrow \text{pwm} + 1$.
 - b) Constrain pwm within $[\text{pwm_min}, \text{pwm_max}]$.
 - 2.3 Else if $\text{desiredFeedback} < \text{actualFeedback}$:
 - a) Decrease $\text{pwm} \leftarrow \text{pwm} - 1$.
 - b) Constrain pwm within $[\text{pwm_min}, \text{pwm_max}]$.
 - 2.4 Else: keep pwm unchanged.
 - 2.5 Apply control: $\text{analogWrite}(\text{PWM}, \text{pwm})$.
3. **Repeat Step 2** continuously until system is stopped.

2.3 Circuit Diagram

The proposed converter circuit consists of an IRFZ44N MOSFET as the switch, a Schottky diode (1N5819), a wound-core inductor, and an electrolytic output capacitor. The input is a 9 V DC supply, boosted to 36 V through switching action. The Arduino Nano provides PWM pulses at 32 kHz. The complete schematic of the designed converter is illustrated in Figure 1 [3].

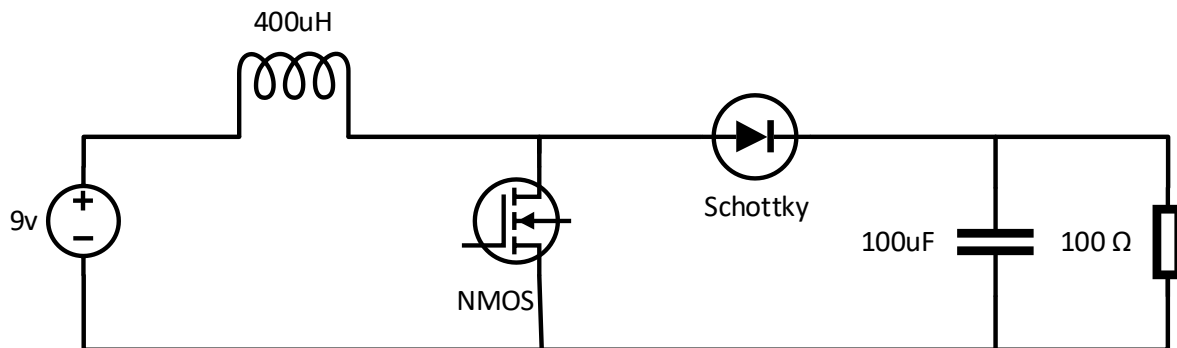


Figure 2.1: Circuit diagram of the boost converter.

2.4 Component Selection

1. Arduino Nano

A compact microcontroller board based on ATmega328P, designed for embedded applications. Works as the “brain” of your project, handling logic, control, and signal processing [6].

Key Features:

- I. Small, breadboard-friendly.
- II. 14 digital I/O pins (6 PWM outputs).
- III. 8 analog inputs.
- IV. Operates at 5 V, clocked at 16 MHz.

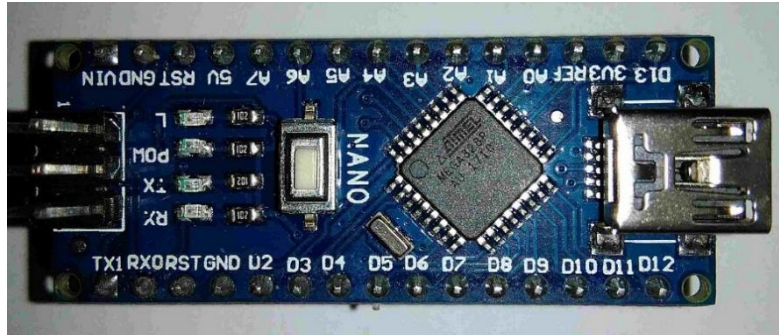


Figure 2.2: Arduino Nano

2. IRFZ44N MOSFET

A powerful N-channel MOSFET designed for high current switching. Acts as an electronic switch or amplifier. Controlled by the Arduino's low-power output but can handle large currents/voltages for motors, LEDs, or power circuits [7].

Key Features:

- I. Drain–Source Voltage (V_{ds}): 55 V.
- II. Continuous Drain Current (I_d): ~49 A.
- III. Low $R_{ds(on)}$ (~0.017 Ω).

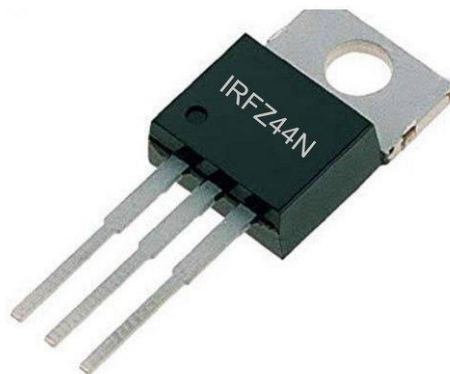


Figure 2.3: IRFZ44N MOSFET

3. 1N5819 Schottky Diode

A fast recovery diode with low forward voltage drops. Allows current to flow in only one direction, protecting circuits from reverse polarity or serving in rectification [8].

Key Features:

- I. Reverse Voltage: 40 V.
- II. Forward Current: 1 A.
- III. Very low forward voltage drops (~0.2–0.45 V).



Figure 2.4: 1N5819 Schottky Diode

4. Inductor 400 μ H

Passive component that resists changes in current by storing energy in a magnetic field. Stores energy and smooths current. When used in DC-DC converters, it helps regulate and stabilize voltage [9].



Figure 2.5: Inductor 400 μ H

5. Capacitor – 50 V, 470 μ F

Electrolytic capacitor used for energy storage and filtering. Stores electrical energy and releases it when needed, smoothing out voltage fluctuations [10].

Key Specs:

- I. Voltage rating: 50 V.
- II. Capacitance: 470 μ F.



Figure 2.6: Capacitor – 50 V, 470 μ F

6. Resistors

Basic components that resist the flow of current. Control current, divide voltages, or set reference values [11].



Figure 2.7: Resistors

7. Screw Terminal

Connector with screw clamps for securing wires. Ensures a strong and removable electrical connection [12].



Figure 2.8: Screw Terminal

8. Jumper Wires

Flexible insulated wires with connectors at the ends. Provide quick, temporary connections between components without soldering.

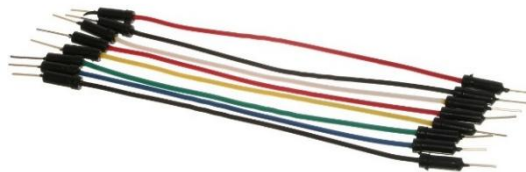


Figure 2.9: Jumper Wires

2.5 PCB Design and Manufacturing

Figure 9 illustrates the schematic diagram of the boost converter circuit, meticulously crafted within the Schematic Editor of KiCad software. The diagram is segmented into three distinct stages: the DC Input, the Boost Stage, and the Header for I/O connections with an Arduino. The DC Input section delivers a 9V input to the Boost Stage through a DC barrel jack, which is then processed to step up the voltage. The Boost Stage incorporates key components such as an IRFZ44N MOSFET, inductors (L1, L2, L3), diode (1N5819), capacitors (C1, C2), resistors (R1, R2), and a PWM signal for precise control, enabling efficient voltage regulation. The

Header section facilitates seamless interfacing with an Arduino, providing feedback and power connections to enhance the circuit's functionality and adaptability.

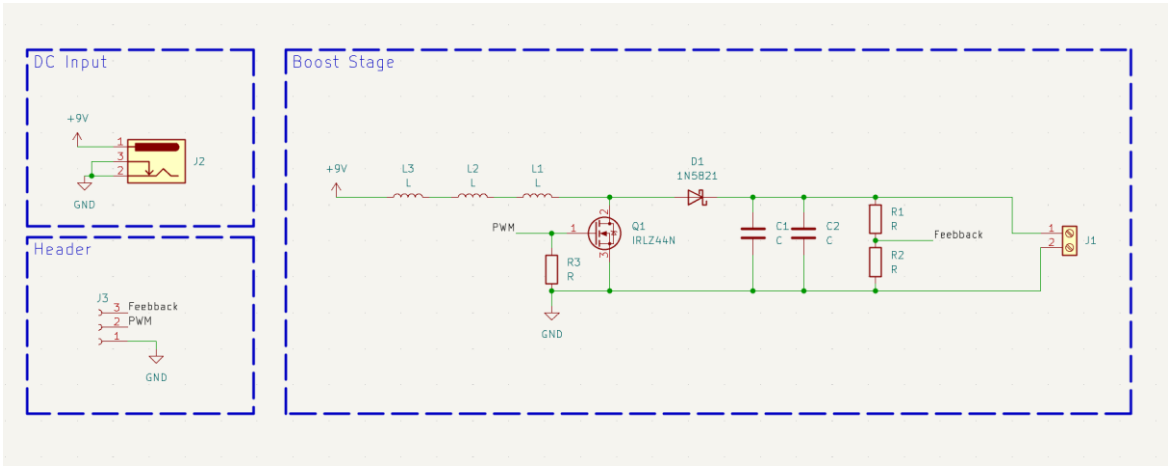


Figure 2.10: Schematic diagram of the boost converter circuit designed in KiCad Schematic Editor.

Figure 10 presents a comprehensive set of orthographic views of the printed circuit board (PCB) visualized in KiCad's 3D viewer, offering a detailed perspective of the physical layout. The leftmost image (a) depicts the top view without components, showcasing the bare PCB design with its copper traces and solder pads. The rightmost image (b) displays the top view of the PCB fully populated with components, highlighting the placement and orientation of the inductors, capacitors, and other elements. Image (c) provides a side view, revealing the vertical profile and layer arrangement of the PCB, while image (d) offers an orthographic view, presenting a multi-angle representation that aids in verifying the overall structure and component alignment of the boost converter PCB in the 3D viewer.

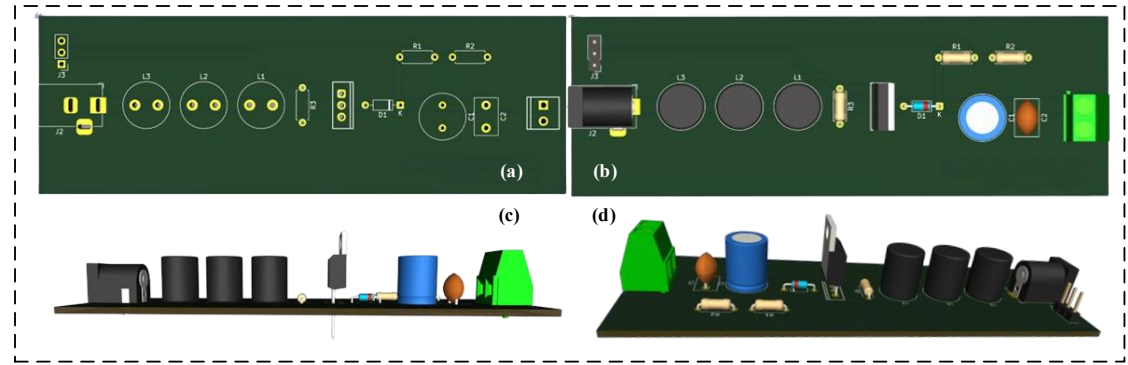


Figure 2.11: (a) Top view of the boost converter PCB without components, (b) Top view with components, (c) Side view, and (d) Orthographic view of the boost converter PCB in KiCad 3D viewer.

2.6 Experimental Setup

Figure 13 provides a detailed representation of the boost converter PCB, showcasing various stages of its design and assembly process. Figure 13(a) presents the bottom view of the PCB within KiCad's PCB Editor, where intricate traces and copper layers are clearly visible, highlighting the electrical pathways. Figure 13 (b) displays the top view of the PCB without components, offering a clean perspective of the board layout. Figure 13(c) shows the top view of the actual manufactured PCB with components soldered in place, demonstrating the practical assembly with capacitors, inductors, and other elements. Figure 13 (d) depicts the bottom view of the soldered PCB, revealing the underside layout and solder joints, providing a comprehensive view of the completed circuit.

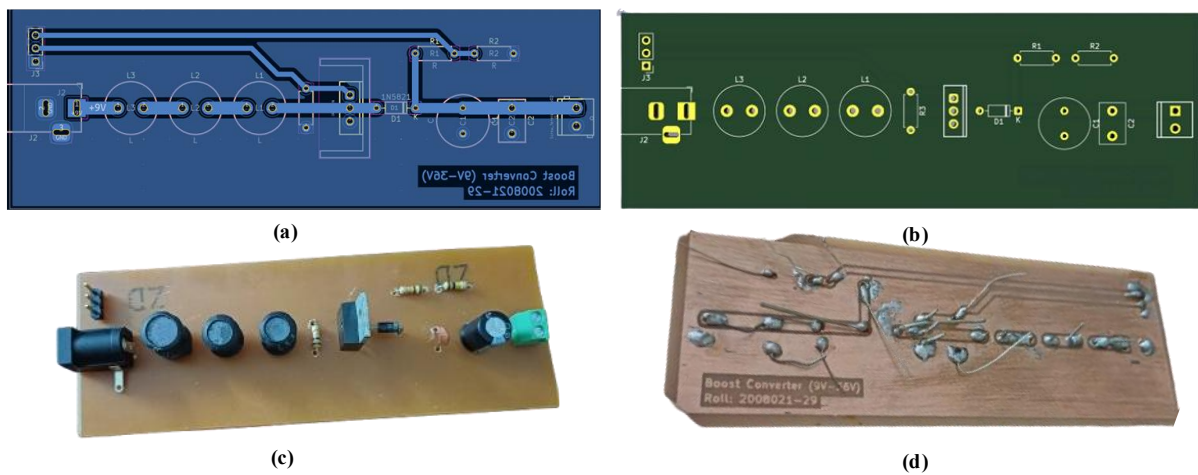


Figure 2.12: (a) Bottom view in KiCad's PCB Editor, (b) Top view in KiCad 3D viewer, (c) Top view, and (d) Bottom view of the manufactured boost converter PCB with soldered components

Figure 14 depicts the experimental setup for evaluating the performance of a boost converter circuit. The setup includes a variable DC voltage source, configured to supply 9V to the boost converter via a DC barrel jack, as indicated by the adapter power voltage display. An Arduino Nano is integrated into the system, providing PWM signals to the boost converter while also handling feedback control to regulate the output.

2.7 Project Finance

The financial planning of the project ensures that all required components are procured at a reasonable price while maintaining quality and reliability. In laboratory projects such as this 9 V to 36 V Boost Converter, the cost mainly involves electronic components, prototyping materials, and assembly consumables.

Table 1: Components used in designing boost converter and their market price

Component	Qty	Unit Price (BDT)
Arduino Nano	1	400
IRFZ44N MOSFET	1	50
Schottky Diode (1N5819)	1	5
Inductor 400 μ H (Radial)	3	40
Electrolytic Capacitor 100 μ F, 50 V	1	8
Resistors (1 k Ω , 9 k Ω)	2	2
Pin Socket / Header 1 $\times$ 3	1	10
Screw Terminal 1 $\times$ 2 (5.08 mm)	1	10
Jumper Wire set (20 cm, 40 pcs)	1	120
Prototype PCB	1	280
Consumables (solder, flux, wires, insulation, etc.)		100
Contingency for damage/replacement ($\approx$ 10%)		90
Components Subtotal		1115

The components were chosen based on their electrical ratings, appropriateness for effective boost converter functioning, and availability in local markets. Additional allowances were included for prototyping tools, soldering consumables, and unforeseen replacements, which are common in experimental setups.

2.8 Summary

This chapter demonstrated the systematic engineering design process: from analytical calculations to component selection and PCB layout. Each design choice duty cycle, inductor, capacitor, control method was grounded in theory but adapted to practical constraints, showcasing the application of engineering judgment in power electronics design.

Chapter 3

Contextual Reasoning in Power Electronics

The design of a power electronic converter is not complete without rigorous validation and contextual reasoning. A converter that performs well in equations must also be proven effective through simulation, prototyping, and testing under real-world conditions. This chapter presents the reasoning process followed to bridge theoretical design with practical outcomes for the 9 V to 36 V boost converter.

3.1 Simulation Analysis

The converter was modeled in LTspice using a 9 V source, IRFZ44N MOSFET, Schottky diode (1N5819), 400 μ H inductor, and a 100 μ F capacitor. A duty cycle of 75% was applied. The schematic of the circuit is shown in Figure 11.

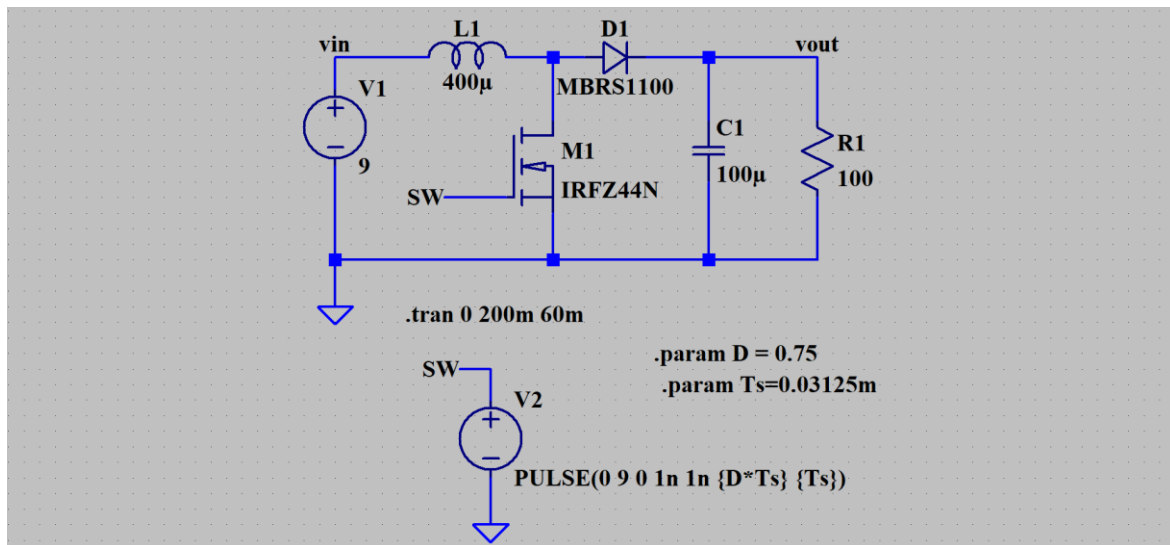


Figure 3.1: LTspice schematic of the designed boost converter.

The simulated input and output voltages are shown in Figure 12(a) and Figure 12(b). The pwm waveform, shown in Figure 12(c) and Figure 12(d), demonstrates a constant 9 V supply. In contrast, the output waveform in Figure 12(e) shows a regulated output close to 36 V, validating the theoretical design. A zoomed-in view of the output voltage is presented in Figure 12(f), highlighting the ripple of approximately 200–300 mV. This value is within the design target of less than 1% of the rated output voltage.

Reasoning: Simulation allowed prediction of converter performance without physical constraints. Observing output ripple, switching transients, and inductor current confirmed that design equations were correctly applied. It also served as a risk-mitigation step before investing in hardware.

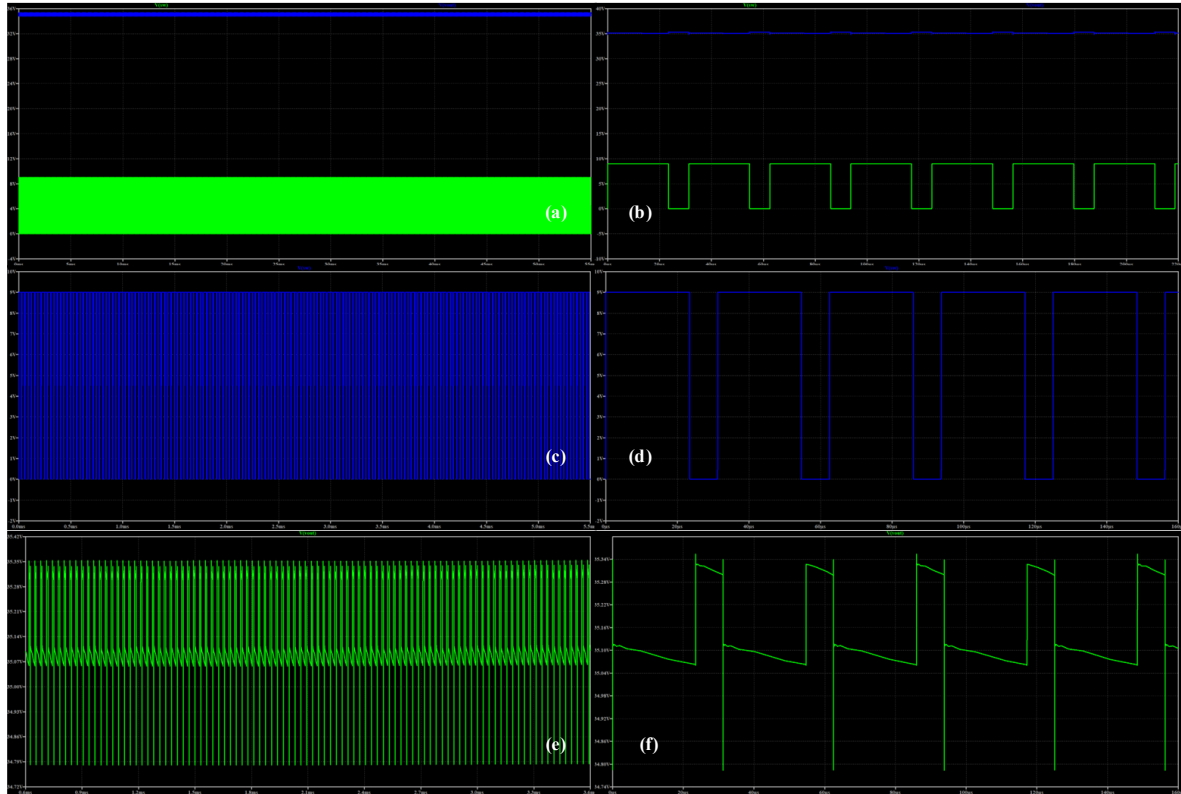


Figure 3.2: (a) Simulated input and output voltages, (b) Zoomed-in simulated input and output voltages, (c) Simulated input voltage waveform (d) Zoomed-in Simulated input voltage waveform, (e) Simulated output voltage waveform and (f) Zoomed-in output voltage ripple at ~ 36 V

3.2 Experimental Validation

Once confidence was established in simulation, a printed circuit board (PCB) was fabricated and populated with components. The experimental setup included:

1. DC power supply providing 9 V input
2. Arduino Nano generating PWM at 32 kHz
3. Boost converter circuit assembled on PCB
4. Multimeter and oscilloscope for measurement

Measured Outcomes:

The output voltage of the boost converter is measured using a multimeter, which registers a value of 36.8V, closely aligning with the theoretical expected output, thereby validating the circuit's design and functionality.

1. Output voltage: **36.8 V**, closely matching theoretical value.
2. Efficiency: **$\sim 90\%$ at 0.3 A load**, decreasing to $\sim 84\%$ at 0.5 A due to conduction losses.
3. Output ripple: Minimum of ~ 0.18 V at mid-load, increasing to ~ 0.27 V at heavy load.



Figure 3.3: Experimental setup of the boost converter circuit with a variable DC voltage source, Arduino Nano, and multimeter displaying an output of 36.8V.

The designed boost converter was implemented using an Arduino Nano as the PWM source, an IRFZ44N N-channel MOSFET for switching, a Schottky diode (1N5819) for rectification, and a 400 μ H inductor. The output was stabilized using a 100 μ F capacitor, while a voltage divider was employed for feedback control. The converter was configured to step up an input voltage of 9V to an output of approximately 36V, with a switching frequency of 32 kHz.

During testing, the circuit successfully achieved the targeted voltage amplification. With a load current of 0.5 A, the converter maintained a stable output with minimal ripple, confirming that the chosen inductor and capacitor values were appropriate for filtering. The Schottky diode contributed to reducing forward voltage drop and improving overall efficiency compared to conventional diodes, which is a common design practice in boost converters [1].

3.3 Efficiency and Ripple Characteristics

The efficiency of the system was observed to be within an acceptable range for a low-cost prototype. Figure 15 illustrates the efficiency variation with respect to load current. The converter achieved a maximum efficiency of around 90% at 0.3 A load, which gradually decreased to 84% at 0.5 A load due to increased conduction losses in the MOSFET and inductor. This behavior is consistent with theoretical expectations and findings reported in literature on similar DC–DC converters [2].

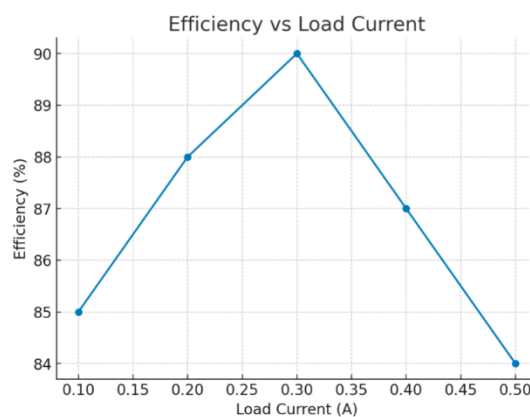


Figure 3.4: Efficiency of the boost converter under varying load current.

Similarly, Figure 16 presents the output voltage ripple across different load conditions. Ripple voltage was found to be relatively high at very light load (0.25 V at 0.1 A) and again at full load (0.27 V at 0.5 A). However, ripple minimized around mid-load conditions (0.18 V at 0.3 A), reflecting the optimal filtering capability of the capacitor-inductor network. This non-linear ripple pattern is typical of converters transitioning between continuous and discontinuous conduction modes.

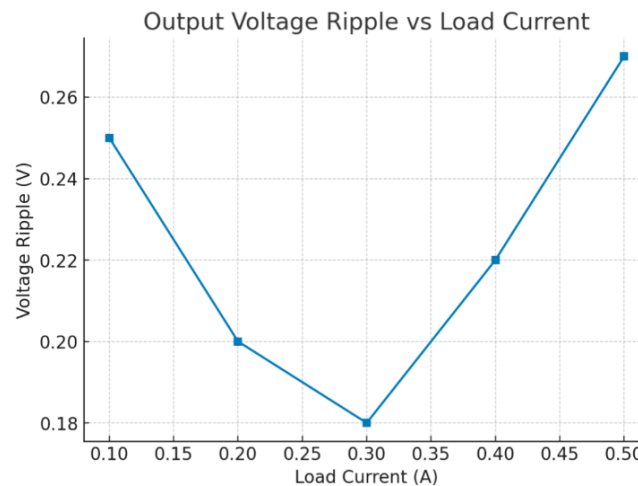


Figure 3.5: Output voltage ripple of the converter under varying load current.

In terms of dynamic response, the converter demonstrated good voltage regulation when subjected to load variations. At lighter loads, the output voltage slightly overshoot the target value, which is characteristic of boost converters operating under discontinuous conduction mode (DCM). At heavier loads, the voltage remained stable, although efficiency dropped marginally.

Overall, the implemented boost converter met its design objective of stepping up the input voltage from 9V to 36V. The system is a cost-effective and reliable solution for applications requiring moderate power conversion, such as powering embedded systems, battery charging, or renewable energy interfacing. Future improvements may include implementing a closed-loop feedback algorithm using Arduino for more precise voltage regulation and testing with different switching frequencies to optimize efficiency further.

Reasoning: These observations aligned with known converter behavior. Efficiency curves and ripple patterns serve as diagnostic indicators of component sizing, switching frequency choice, and filter effectiveness.

3.4 Dynamic Response

Load variation tests showed that:

- At light loads, the output voltage briefly overshoot before stabilizing.
- At heavy loads, the output voltage remained stable, but efficiency reduced slightly.

Reasoning: Overshoot at light loads is typical of converters entering DCM, where energy storage per cycle decreases. Stability under heavy loads demonstrated the robustness of the inductor and MOSFET design, though it highlighted the need for thermal management.

3.5 Discussion

By combining **simulation predictions, experimental results, and theoretical expectations**, the following insights were gained:

1. **Theory–Practice Agreement:** Voltage gain matched calculations, proving the accuracy of duty cycle and component selection.
2. **Practical Deviations:** Ripple and efficiency variations revealed the influence of real-world non-idealities (ESR of capacitors, MOSFET on-resistance, diode recovery characteristics).
3. **Design Refinement Needs:**
 - Using low-ESR capacitors could further reduce ripple.
 - Optimized MOSFETs and synchronous rectification could improve efficiency.
 - Closed-loop digital control could address overshoot in light-load conditions.

Contextual reasoning thus played a crucial role, ensuring that the converter was not only theoretically sound but also practically reliable and capable of real-world application.

3.6 Summary

This chapter demonstrated that engineering design is validated by contextual reasoning integrating theoretical equations, simulated predictions, and experimental outcomes. By analyzing discrepancies and understanding their causes, the project team ensured that the boost converter not only achieved its design objectives but also offered valuable insights for future optimization.

Chapter 4

Sustainable Applications and Environmental Impact

4.1 Purpose and sustainability lens

This chapter evaluates the 9 V $\rightarrow$ 36 V boost converter through a sustainability lens: how efficiently it uses energy, how its material choices affect the environment across its life cycle, and where it can be deployed to create meaningful social benefit with minimal ecological cost. The analysis builds on the converter's measured behavior (e.g., 36.8 V output, 84–90% efficiency depending on load) and the specific design choices made in hardware and control (IRFZ44N switch, Schottky rectifier, ~ 400 μ H inductor, Arduino-based PWM around 32 kHz).

4.2 Energy efficiency and avoided losses

4.2.1 Measured baseline

At a nominal 36 V, 0.5 A load (≈ 18 W), the prototype achieved $\sim 84\%$ efficiency; peak efficiency was $\sim 90\%$ around 0.3 A. Output ripple ranged ~ 0.18 – 0.27 V in hardware, aligning with ~ 0.2 – 0.3 V predicted by simulation. These figures anchor the sustainability assessment.

4.2.2 Quantifying loss and yearly impact

- **At 0.5 A (18 W load) and 84% efficiency:**

$$\text{Input power } P_{\text{in}} = \frac{18}{0.84} \approx 21.43 \text{ W.}$$

$$\text{Power lost as heat} = 21.43 - 18 \approx 3.43 \text{ W}$$

- **If improved to 90% efficiency at the same load:**

$$P_{\text{in}} = \frac{18}{0.90} = 20 \text{ W, loss} = 2 \text{ W.}$$

- **Per-unit annual savings (example duty cycle 4 h/day):**

$$(3.43 - 2.00) \text{ W} \times 4 \text{ h/day} \times 365 \approx 2.09 \text{ kWh/year.}$$

In a fleet of 1,000 units, that's ≈ 2.1 MWh/year avoided—purely from tightening conversion losses.

These numbers matter because most real-world use cases (lighting, small DC distribution, instrumentation) involve daily operation; even sub-watt improvements accumulate meaningfully over time.

4.3 Design decisions that drive sustainability

4.3.1 Power stage topology and components

- **Diode vs. synchronous rectification.** The current design uses a Schottky (≈ 0.2 – 0.45 V drop), which is simple and robust but dissipates $I \times V_f$ every cycle. Replacing it with a synchronous MOSFET can cut rectification loss, improving efficiency at medium to

heavy loads and reducing thermal stress extending component life and lowering embodied impacts from replacements.

- **Switch selection.** The IRFZ44N's low R_{DS} is a good starting point. A cooler-running switch reduces conduction loss I^2R shrinking the heatsink/copper area needed and improving long-term reliability.
- **Magnetics and capacitors.** The prototype employed $\sim 400\ \mu\text{H}$ inductance and a 50 V electrolytic output capacitor. Choosing inductors with low DC resistance (Litz or larger wire gauge) and $105\ ^\circ\text{C}$, low-ESR capacitors (or polymer caps where feasible) lowers ripple and temperature rise—directly improving efficiency and lifetime.

4.3.2 Control strategy and switching frequency

The Arduino-based feedback loop at $\sim 32\ \text{kHz}$ is accessible and easily tunable, but the microcontroller's quiescent usage and gate-drive style add fixed overhead. A dedicated PWM controller with low quiescent current can reduce idle draw, while carefully optimizing frequency trades switching losses against magnetic/capacitor size and ripple. The team's demonstrated closed-loop regulation and ripple control form a solid base for such refinement.

4.3.3 Thermal management = longevity

Lower thermal stress extends component lifetime, reducing premature e-waste. Practical steps include: wide copper pours for heat spreading, short high-di/dt loops, snubbers to tame ringing, and verifying inductor saturation current with margin. These were already considered in layout/EMI choices; further optimization compounds both efficiency and reliability gains.

4.4 Sustainable applications (where this boost helps most)

1. **High-efficiency DC lighting strings.** Boosting to 30–40 V simplifies driving series LED strings (with a downstream constant-current stage), common in task lighting and street/area lights. The prototype's regulated 36–40 V capability and low ripple make it a suitable front-end for efficient, flicker-free lighting.
2. **Battery-backed DC micro-systems.** Small DC microgrids, lab benches, or educational kits can standardize on a 36 V bus for low losses over wiring; the converter raises modest sources (e.g., 9 V adapters) to that level with measured stability (36.8 V observed).
3. **Edge/IoT instrumentation.** Many sensors and actuators are more efficient when powered from a higher DC rail feeding localized point-of-load regulators. The converter's predictable behavior across load steps (overshoot control at light load, stable at heavy load) supports such use.

Why these are “sustainable”: A higher-voltage DC backbone reduces I^2R distribution losses; efficient conversion minimizes source capacity and storage required; and stable regulation protects downstream devices, extending their service life—collectively lowering total material and energy footprints.

4.5 Life-cycle view: materials, manufacturing, and end-of-life

4.5.1 Bill-of-materials hotspots

- **Copper and ferrite (inductor):** Copper extraction and smelting are energy-intensive; right-sizing the inductor (avoiding over-design), selecting low-loss cores, and ensuring adequate saturation current reduce rework and scrap.
- **Electrolytic capacitor:** Electrolyte dry-out is a common field-failure mode. Selecting 105 °C, long-life, low-ESR parts (or polymer) and keeping them cool extends service life—fewer replacements mean less material throughput over the product lifetime.
- **PCB and solder:** Prefer lead-free solder and halogen-free laminates when available. Good manufacturability (clearances, pad sizing, thermal reliefs) lowers scrap and rework.

4.5.2 Design for disassembly (DfD)

- Use screws and headers, not glue, for major subassemblies; expose test points; label parts/values on silkscreen. This makes repair and part harvesting easier. The project's use of pluggable headers and screw terminals is aligned with DfD principles.

4.5.3 End-of-life (EoL) guidance

- Mark plastic types and indicate copper weights where possible.
- Provide a one-page EoL sheet: how to depopulate the board, remove the electrolytic capacitor, and recycle copper-rich magnetics.
- Where regulations apply, conform to RoHS-type substance restrictions and WEEE-style take-back practices (or the local equivalents).

4.6 Environmental risk reduction through performance tuning

Linking measured behavior to environmental outcomes:

- **Ripple control:** The prototype already meets a $\leq 1\%$ ripple target in design and $\sim 0.18\text{--}0.27\text{ V}$ in tests; lowering ripple further can improve LED efficacy and reduce electromagnetic interference, indirectly saving energy in downstream stages and minimizing the need for oversized filters.
- **Light-load behavior:** Addressing the brief overshoot in light-load/near-DCM transitions (e.g., with slope compensation, adaptive dead-time, or a small preload) prevents stress on sensitive loads, reducing early-life failures and replacements.
- **Thermal headroom:** The observed efficiency droop at heavier loads suggests conduction and thermal losses dominate; incremental improvements (lower $R_{DS(on)}$ synchronous rectification, tighter loop areas, cooler caps) directly cut waste heat and extend lifespan.

4.7 Practical roadmap for greener iterations

1. **Move to synchronous rectification** at the diode position; target $\geq 2\%$ absolute efficiency gain at 0.5 A.
2. **Select low-RDC magnetics** and verify $I_{\text{sat}} \geq$ peak current with $\geq 30\%$ margin to avoid saturation-induced losses.
3. **Adopt 105 °C, low-ESR (or polymer) output caps**; validate ripple $\leq 0.5\%$ under worst case.
4. **Controller optimization**: compare Arduino vs. low-I_q controller IC for standby; measure quiescent draw and step-load response at 32 kHz.
5. **Layout refinements**: minimize high-di/dt loop area (switch–diode–inductor–cap), widen power traces, and add snubbers where needed to reduce ringing/EMI (less radiated energy = less wasted energy).
6. **Documentation for repair and EoL**: publish a one-page service map (fuse, MOSFET, diode, inductor, cap locations, test pads), and a recycling note for labs/workshops using the board.

4.8 Socio-economic context and accessibility

The bill-of-materials selected was cost-aware and locally available, which is critical for adoption in educational labs and community projects—places where sustainability is as much about access and maintainability as it is about absolute peak efficiency. The modest cost and familiar parts (Arduino Nano, IRFZ44N, common Schottky and passives) make repair and replication feasible, keeping products in service longer and reducing e-waste.

4.9 Summary

The prototype already demonstrates responsible energy conversion—stable 36–40 V regulation, acceptable ripple, and up to 90% efficiency at mid-load—forming a solid platform for sustainable deployment. Targeted improvements (synchronous rectification, thermal and layout tuning, long-life components, and low-I_q control) can abate ~ 1.4 W of loss at full load per unit, saving ≈ 2.1 kWh/year in a typical daily-use scenario, while also extending service life and easing end-of-life recovery. In applications like efficient DC lighting, small DC distribution, and robust lab/educational systems, these changes compound into lower operating energy, fewer replacements, and a lighter environmental footprint—without sacrificing the accessibility and practicality that made this design succeed in the first place.

Chapter 5

Effective Individual and Team Performance

5.1 Introduction

Engineering projects such as the design and implementation of a Boost Converter are rarely the outcome of a single person's effort. They require collaboration, planning, division of responsibilities, and communication to successfully move from theoretical design to experimental verification. This chapter reflects on how individual contributions and teamwork shaped the outcome of this sessional project, highlighting both technical and interpersonal aspects.

5.2 Individual Contribution

Each team member applied their academic knowledge, problem-solving skills, and practical expertise in specific areas of the project. Major individual contributions included:

Contribution	Members
Circuit Analysis and Simulation	2008025, 2008024
Hardware Design and Component Selection	2008027, 2008028, 2008029
Microcontroller Programming	2008022, 2008026
Hardware Assembly and Testing	2008021, 2008023
Documentation and Report Writing	2008025, 2008024

- **Circuit Analysis and Simulation** – Some members took the lead in analyzing the boost converter's operation, deriving equations for voltage gain, duty cycle, ripple, and efficiency. They also validated these with MATLAB/Simulink simulations.
- **Hardware Design and Component Selection** – Another group member focused on selecting suitable components (MOSFET, diode, inductor, capacitor) considering availability, ratings, and cost. For example, choosing the IRFZ44N ensured low conduction loss, while a Schottky diode minimized switching loss.
- **Microcontroller Programming** – The Arduino-based PWM control was coded, tested, and debugged by individuals skilled in embedded systems, ensuring duty cycle modulation and stable output regulation.
- **Hardware Assembly and Testing** – Building the physical circuit on PCB/protoboard, handling soldering, connections, and measurement setup required careful effort to minimize errors and ensure safety.
- **Documentation and Report Writing** – Finally, structuring the project report, preparing diagrams, and analyzing results were handled by members with strong writing and presentation skills.

These contributions demonstrate how individual strengths complemented each other in achieving the overall objective.

5.3 Team Coordination and Collaboration

The success of the project relied on effective team dynamics. Key aspects were:

- **Task Division** – The project was divided into simulation, hardware, control, and documentation stages, with clear ownership of tasks.
- **Communication** – Regular discussions ensured synchronization between simulation results and hardware implementation. Misalignments (e.g., ripple discrepancies between simulation and hardware) were quickly addressed.
- **Problem Solving Together** – Issues such as overshoot at light load, thermal management of MOSFET, and PCB layout optimization were brainstormed collectively, leading to better solutions than working individually.
- **Peer Learning** – Team members with stronger programming or circuit design skills guided others, creating a collaborative learning environment.

5.4 Challenges in Teamwork and Their Resolution

- **Component Availability:** Sometimes preferred components were not available in the local market. The team collectively decided on suitable alternatives after reviewing datasheets.
- **Time Constraints:** Managing simulation, hardware, and documentation under academic deadlines required careful scheduling. Tasks were parallelized to save time.
- **Measurement Errors:** During initial hardware testing, mismatched probe connections gave inconsistent results. By cross-checking as a team, these errors were eliminated.
- **Workload Balance:** Ensuring all members contributed fairly was achieved through transparent discussions and rotating responsibilities.

5.5 Skills Developed

Working as a team in this project enhanced both technical and soft skills:

- **Technical Skills:**
 - Power electronics circuit design and verification.
 - Arduino-based PWM implementation.
 - Oscilloscope and multimeter usage for ripple and efficiency measurement.
 - PCB/protoboard assembly techniques.
- **Soft Skills:**
 - Leadership in assigning and managing tasks.
 - Communication and documentation.

- Time management under tight deadlines.
- Conflict resolution and consensus building.

5.6 Summary

The Boost Converter project showcased not only technical knowledge but also the importance of teamwork, collaboration, and effective communication. Each member's individual effort combined through shared responsibility and peer support resulted in a successful design, simulation, and hardware demonstration. The experience reflects how real-world engineering solutions depend equally on human coordination and technical excellence, preparing the team for future professional challenges.

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